

Chess Tournament Player Management System

Python OOP Solution According to the Given Operations

Complete Solution

```
class ChessTournamentSystem:

    def __init__(self):
        self.players = {}

    # Operation 1: Add Player
    def add_player(self, player_id: str, name: str, rating: int) -> dict:
        if player_id in self.players:
            raise ValueError("Player already exists")

        self.players[player_id] = {
            "name": name,
            "rating": rating,
            "status": "Active"
        }

        return self.players

    # Operation 2: Update Rating
    def update_rating(self, player_id: str, new_rating: int) -> dict:
        if player_id not in self.players:
            raise KeyError("Player not found")

        self.players[player_id]["rating"] = new_rating

        return self.players

    # Operation 3: Get Player Details
    def get_player_details(self, player_id: str) -> dict:
        if player_id not in self.players:
            raise KeyError("Player not found")

        return self.players[player_id]

    # Operation 4: List Qualified Players
    def qualified_players(self, minimum_rating: int) -> list:
        qualified = []

        for player_id, details in self.players.items():
            if details["rating"] >= minimum_rating:
                qualified.append(player_id)

        return qualified
```

Sample Execution

```
chess = ChessTournamentSystem()

print(chess.add_player("P101", "Arjun", 1850))
print(chess.add_player("P102", "Riya", 1700))
print(chess.add_player("P103", "Karthik", 2100))

print(chess.update_rating("P101", 1920))
print(chess.get_player_details("P101"))
print(chess.qualified_players(1800))
```

Expected Final Output

```
{'name': 'Arjun', 'rating': 1920, 'status': 'Active'}  
['P101', 'P103']
```

Error Handling

```
chess.add_player("P101", "Another Player", 1600)  
# ValueError: Player already exists
```

```
chess.update_rating("P999", 2000)  
# KeyError: Player not found
```

```
chess.get_player_details("P999")  
# KeyError: Player not found
```